Python
Week 2

Numbers, Conditionals, & Basic Logic

Objective: Making programs think

What You'll Learn:

- Numbers (int, float)
- Type conversion (int(), str())
- If, elif, else
- Comparison operators

Do Now (Week 1 Review)

Write a program that:

1. Asks for the user's name
2. Creates a variable for the user's favorite food.

```
name = input("What is your name? ")
favorite_food = "pizza"

print("Hi " + name + "! I like " + favorite_food + " too!")
```

Part 1: Numbers

Numbers in Python

Python can work in numbers!

There are two main types:

- **int** -> whole numbers (1, 5, 10)
- **float** -> decimals (3.5, 2.7)

There are many **math operators** (you probably have seen this last week).

- ➕ (adding)
- ➖ (subtracting)
- ✖ (multiply)
- ➗ (divide)
- % (remainder)

Try it:

What will the following print?

a)
```
print(6 * 3)
```

b)
```
print(10 % 3)
```

c)
```
print(10 + 5)
```

d)
```
print(10 % 2)
```

Why does this code break?

```
age = input("How old are you? ")
print(age + 1)
```

Answer: the variable **age** is NOT a number, it's a string. Thus, it cannot be used with normal math operators.

To fix the code, convert the string into a int using **int()**.

```
age = int(input("How old are you? "))
print(age + 1)
```

You can also use the function, **str()** to convert numbers into text.

Example:

```
age = 12
print("You are " + str(age))
```

Part 2: Conditionals

So far, our programs follow instructions exactly. But what if we want the program to choose what to do?

Example:

- If it's raining -> bring an umbrella
- If you're hungry -> eat food

What is an **if** statement?

An **if** statement lets your program check something, then decides what to do.

Structure:

```
if condition:
    do something
```

“If something is true, THEN do this”

Step-by-Step Example:

```python
age = 12

if age < 13:
    print("You are a kid")
```

- Is 12 less than 13? -> YES
- So the code runs
- If it were false -> nothing happens

A **condition** is a question with a TRUE or FALSE answer.

Example:

- Is 5 > 3? -> TRUE
- Is 10 < 2? -> FALSE

Computers only understand TRUE or FALSE.

Comparison Operators

- 🟰🟰 (equal to)
- !🟰 (not equal to)
- < (less than)
- > (greater than)
- <🟰 (less than or equal to)
- >🟰 (greater than or equal to)

IMPORTANT:
🟰 means assign,
🟰🟰 means compare

Else means “otherwise”

```
age = int(input("Enter age: "))

if age < 13:
    print("You are a kid")
else:
    print("You are a teenager")
```

- If condition is TRUE -> run first block
- Otherwise -> run else block
- “Else” means everything else

Using elif

```
score = int(input("Enter score: "))

if score >= 90:
    print("A")
elif score >= 80:
    print("B")
else:
    print("Keep trying")
```

- Check first condition.
- If false -> check next
- If none match -> use else

Step-by-Step Thinking

Let's say the score = 85

1. Is 85 >= 90 -> NO
2. Is 85 >= 80 -> YES -> print "B"
3. Stop checking.

```python
score = int(input("Enter score: "))

if score >= 90:
    print("A")
elif score >= 80:
    print("B")
else:
    print("Keep trying")
```

Now it's your turn!

Open up Google Classroom and complete Week 2 Classwork.

Due Apr 5, 2027

Positive, Negative, or Zero

Write a program that:
1. Asks the user for a number.
2. Prints
    - "Positive" if the number is greater than 0.
    - "Negative" if the number is less than 0.
    - "Zero" if the number is 0.

Must use:

- input()
- int()
- if, elif, else

Upload a screenshot of your work to this assignment.

Now it's your turn!

Write a program that:

1. Asks the user for a number.
2. Prints
   a. "Positive" if the number is greater than 0.
   b. "Negative" if the number is less than 0.
   c. "Zero" if the number is 0.

Must use:

   a. input()
   a. int()
   b. if, elif, else

Classwork Answer:

```
number = int(input("Enter a number: "))

if number > 0:
    print("Positive")
elif number < 0:
    print("Negative")
else:
    print("Zero")
```

Thanks!

Have questions? Reach out:

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